

ConDyGNet: Constraint-Guided Dynamic Graph Networks for Multivariate Time Series Forecasting

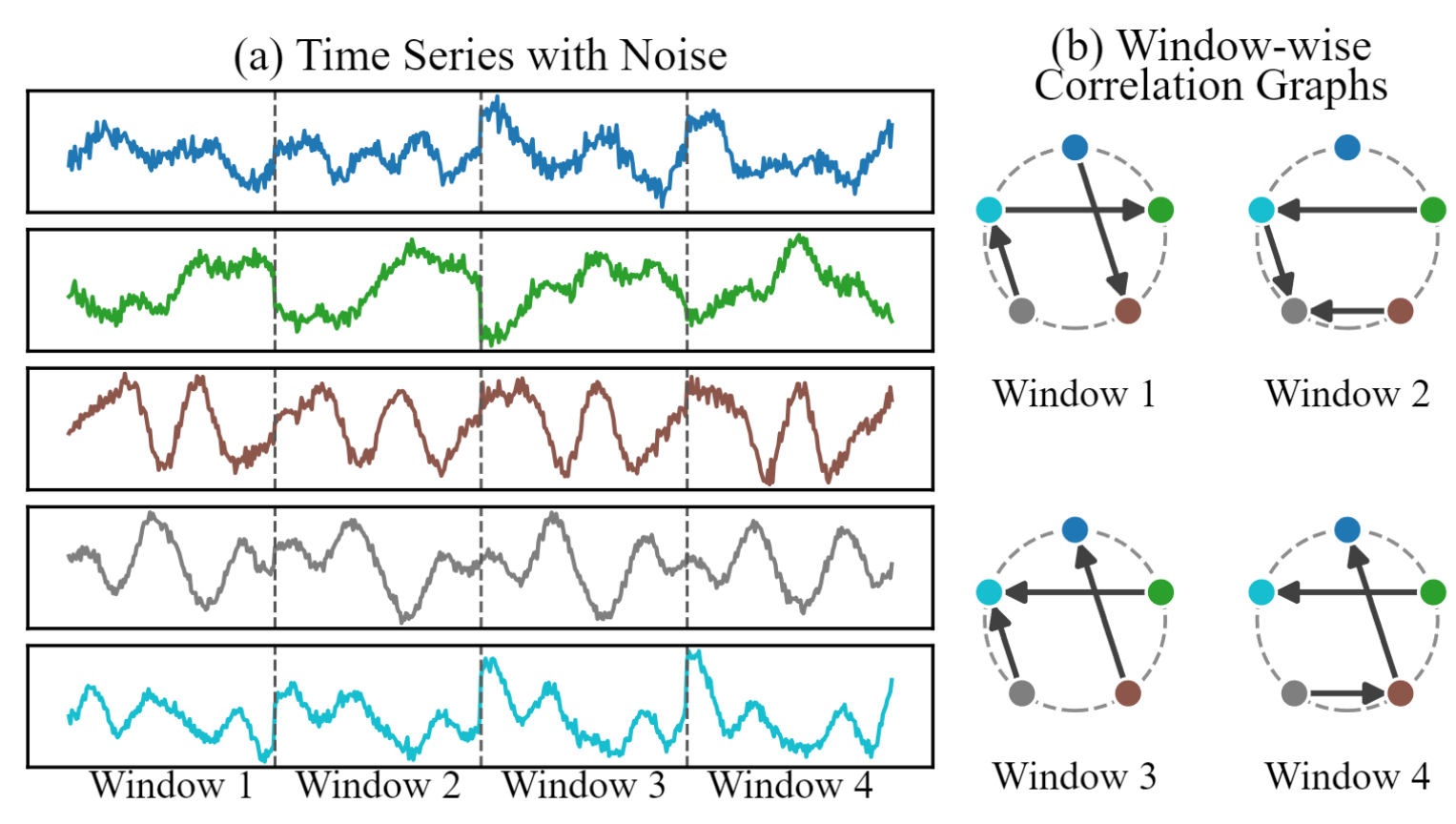
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MOTIVATION

Forecasting depends on relationships between variables, yet those relationships are both time-varying and easily corrupted by short-window noise.



The stability–adaptivity dilemma

A single global graph is stable, but cannot follow changing inter-channel dependencies. A fully dynamic graph adapts locally, but may mistake noise for structure and spread that noise during message passing.

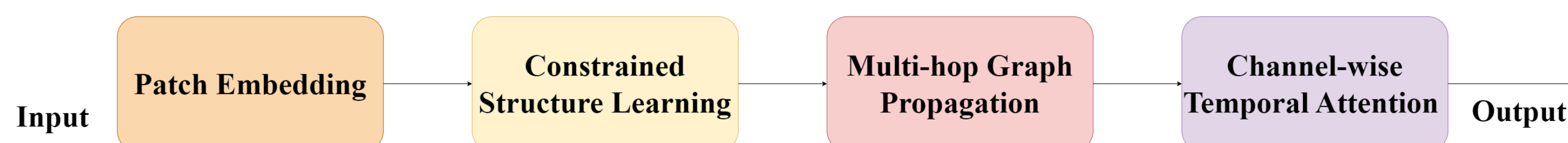
Our idea

ConDyGNet learns a compact global structural vocabulary, then predicts only patch-specific mixing weights. Local graphs can change over time, while remaining inside a reliable global search space.

GLOBAL BASIS · DYNAMIC WEIGHTS

METHODS

ConDyGNet combines constrained graph construction with graph propagation and temporal attention in one end-to-end forecasting model.



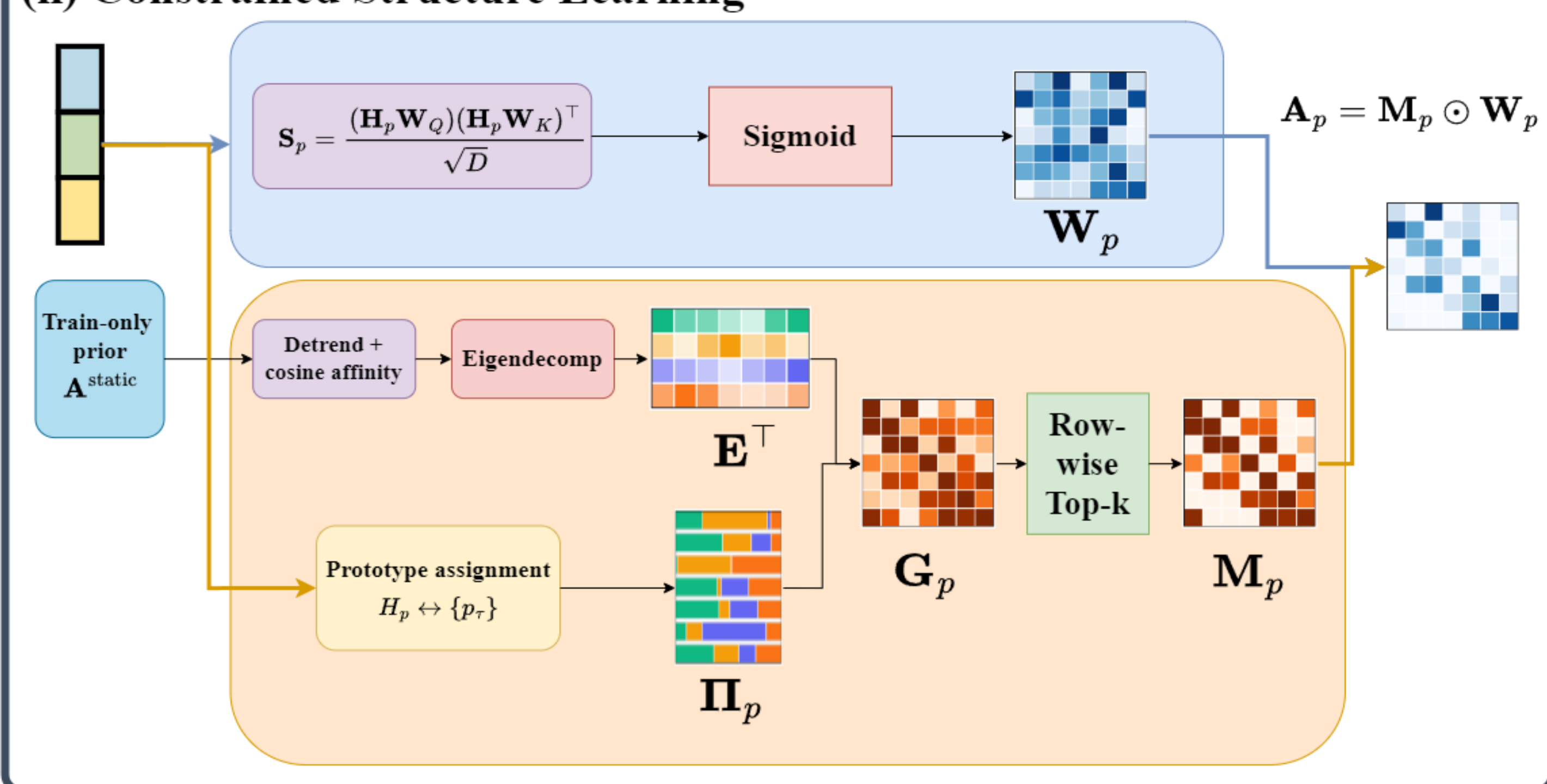
• 1 Patch embedding

Each variable is divided into non-overlapping temporal patches. A shared projection converts every patch into a channel token, preserving local temporal context while keeping channels explicit.

• 2 Global structural prior

Slow trends are removed before measuring residual co-movement. Dominant recurring interaction patterns form a low-dimensional basis bank, which acts as a structural anchor rather than a fixed graph.

(ii) Constrained Structure Learning



• 3 Context-aware dynamic graph

Patch context routes each channel over the shared basis modes. The model keeps only the strongest supported neighbors, then estimates interaction strength on those retained edges. This separates which edges are trustworthy from how strongly they should communicate.

• 4 Graph and temporal forecasting

Multi-hop propagation aggregates information across the constrained graph with a root-state residual. Channel-wise temporal attention then models dependencies across patches before the final multi-horizon projection.

EXPERIMENTS

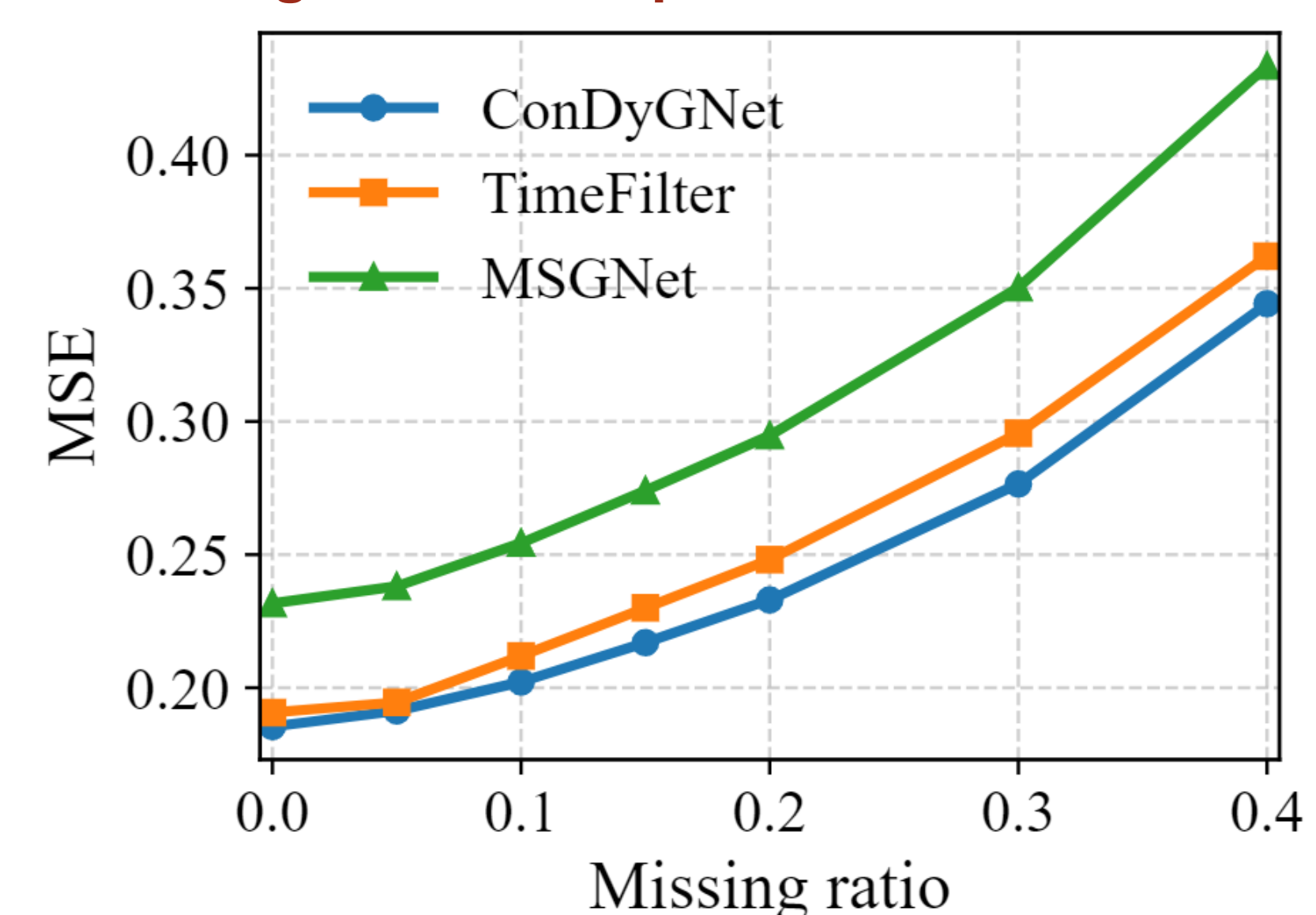
Evaluation covers eight public datasets, four prediction horizons, and ten representative GNN, Transformer, MLP, convolutional, and linear baselines.

MEAN OVER ALL 32 DATASET–HORIZON SETTINGS IN TABLE 1

Method	Architecture family	Mean MSE	Mean MAE
ConDyGNet	GNN + Transformer	0.3057	0.3133
TimeFilter	GNN	0.3076	0.3227
SOFTS	MLP	0.3212	0.3299
TimeXer	Transformer	0.3219	0.3351
DUET	Clustering + attention	0.3244	0.3229
iTransformer	Transformer	0.3284	0.3358
PatchTST	Transformer	0.3529	0.3549
MSGNet	GNN + attention	0.3598	0.3557
TimesNet	Convolution	0.3669	0.3565
DLinear	Linear	0.3998	0.3970

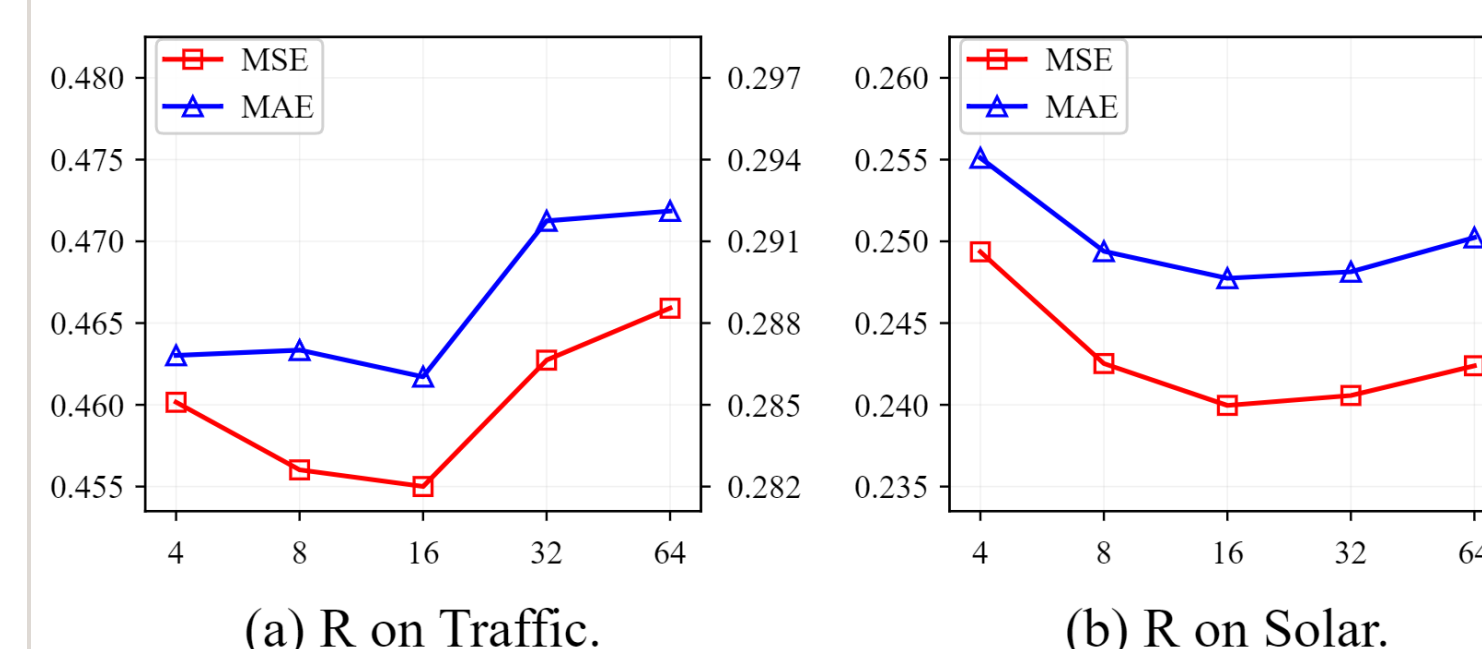
ConDyGNet achieves best or tied-best MAE in 30 settings and best or tied-best MSE in 25. In 25 of 32 settings, it is state of the art on both metrics simultaneously.

Robustness under missing-block corruption



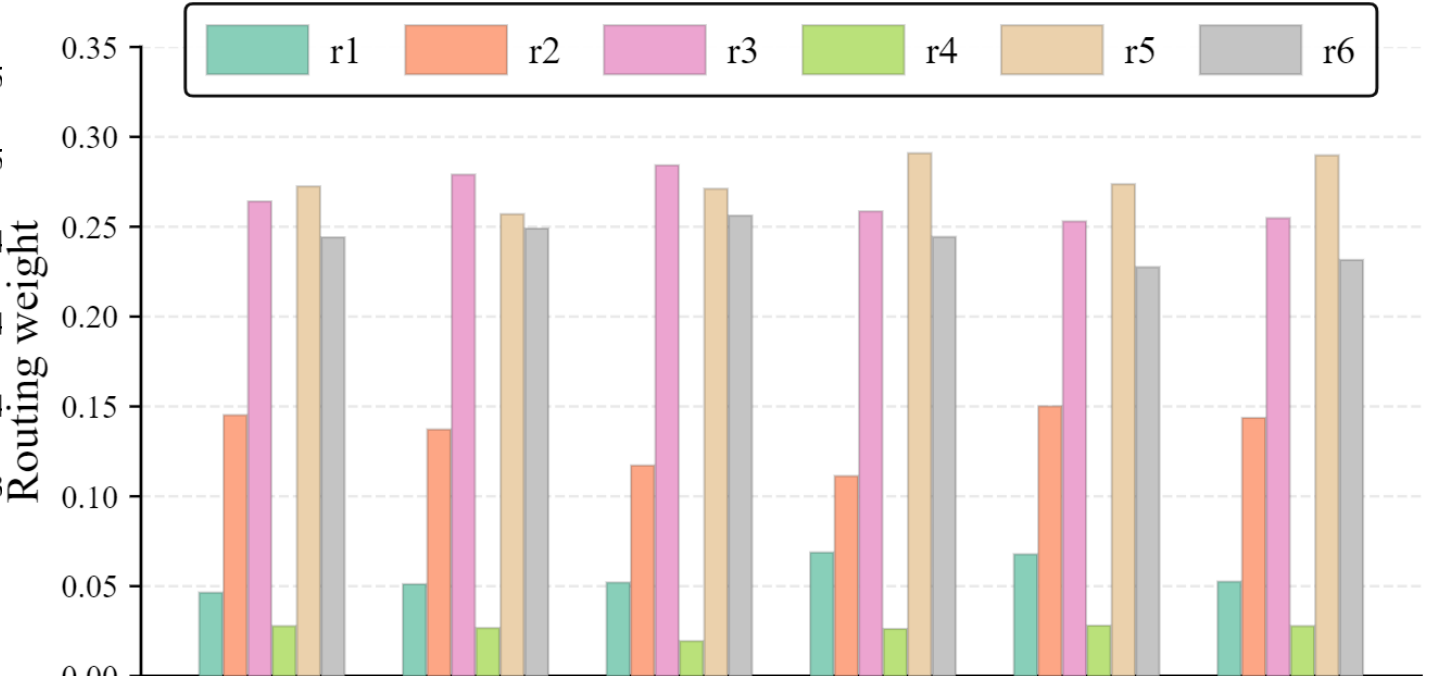
Without retraining, ConDyGNet remains consistently below TimeFilter and MSGNet as missingness increases.

Capacity control



Moderate rank balances expressiveness and structural regularization; very large rank weakens the constraint.

Patch-wise routing



Routing concentrates on a few modes and evolves smoothly between adjacent patches.

The three analyses support the same mechanism: the shared basis limits unstable structural drift, patch routing keeps local adaptation selective, and the constrained graph degrades more slowly when observations are corrupted.

CONCLUSION

Reliable dynamic graphs need both a stable structural vocabulary and local, context-sensitive adaptation.

• Constrained yet dynamic

A global low-dimensional basis captures recurring inter-channel patterns, while patch-specific routing changes the graph as the time-series context evolves.

• Noise-aware communication

Sparse structural support filters suspicious dependencies before propagation; interaction strength is estimated only on the retained edges.

• Consistent empirical gains

Across diverse forecasting architectures and datasets, ConDyGNet improves accuracy and remains robust when test windows contain missing blocks.

ConDyGNet preserves local graph dynamics without giving up global structural reliability.